

Cross-Validated Numerical Predictions of Electromagnetic Stress Asymmetry in Asymmetric Resonant Cavities

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Abstract

We present a cross-validated numerical investigation of electromagnetic stress asymmetry in asymmetric resonant cavity systems. Reduced-order finite-difference time-domain and finite-element-inspired calculations are used to evaluate field-energy distributions, asymmetry parameters, convergence behavior, and effective force scaling under resonant conditions. Parameter sweeps, mesh-convergence tests, and control configurations are included to assess numerical robustness. The results show consistent asymmetry-dependent scaling across methods and define explicit falsifiable predictions for experimental testing, including power scaling, resonance dependence, symmetry suppression, and cavity-orientation sensitivity. These findings provide a reproducible and quantitatively constrained computational benchmark that directly informs experimental validation efforts for evaluating potential electromagnetic stress effects in asymmetric resonant cavity systems.

1 Introduction

Recent theoretical, simulation-based, and experimental-baseline work has explored whether asymmetric electromagnetic field configurations may produce measurable stress imbalances in resonant cavity systems. While phenomenological models and preliminary simulations suggest possible asymmetry-dependent effects, numerical robustness and reproducibility remain critical before such predictions can be meaningfully tested in hardware.

This work presents a cross-validated numerical study designed to test whether predicted asymmetry scaling remains stable across independent computational approximations. The objective is not to claim experimental detection of a physical effect, but to establish whether the numerical predictions are internally consistent, reproducible, and suitable for direct experimental comparison.

2 Model Description

2.1 Geometry

A truncated conical cavity is used to introduce controlled geometric asymmetry:

- Length: $L \sim 10$ cm
- Small radius: $r_1 \sim 2$ cm
- Large radius: $r_2 \sim 5$ cm

A symmetric reference cavity is also simulated as a control case.

2.2 Energy Density and Asymmetry Parameter

The electromagnetic energy density is approximated as

$$\nu = \frac{1}{2}\epsilon_0 E^2 + \frac{1}{2\mu_0} B^2. \quad (1)$$

The dimensionless asymmetry parameter is defined by partitioning the cavity into opposing integration regions. The integration regions are defined by bisecting the cavity along its longitudinal axis, separating the volume into forward and rearward halves relative to the geometric asymmetry:

$$A = \frac{\int(\nu_+ - \nu_-) dV}{\int(\nu_+ + \nu_-) dV}. \quad (2)$$

The effective force scaling is evaluated using

$$F \propto \chi_{\text{eff}} A (E^2 + B^2). \quad (3)$$

This scaling is treated as a phenomenological prediction rather than a derived microscopic force law.

3 Numerical Methods

3.1 FDTD-like Reduced-Order Simulation

A finite-difference time-domain-inspired representation is used to approximate the spatial structure of resonant field-energy distributions. In this context, reduced-order refers to simplified field representations that capture dominant spatial energy distributions without resolving full-wave electromagnetic boundary conditions at high fidelity. The reduced-order grid is not intended to replace a full electromagnetic solver, but it provides a controlled framework for testing asymmetry scaling and convergence behavior.

3.2 FEM-like Frequency-Domain Simulation

An independent finite-element-inspired frequency-domain calculation is used to approximate steady-state field-energy distributions under resonant excitation. Agreement between the two methods is used as a cross-validation criterion.

3.3 Convergence Testing

Mesh refinement and parameter variation are used to assess numerical stability. A result is considered numerically stable when the computed asymmetry parameter changes by less than a few percent under grid refinement and boundary-condition variation.

4 Results

4.1 Field Asymmetry Visualization

Spatial energy-density distributions show asymmetric concentration of field energy within the cavity. The computed map in Fig. 1 gives an asymmetry parameter of approximately $A \approx 0.136$.

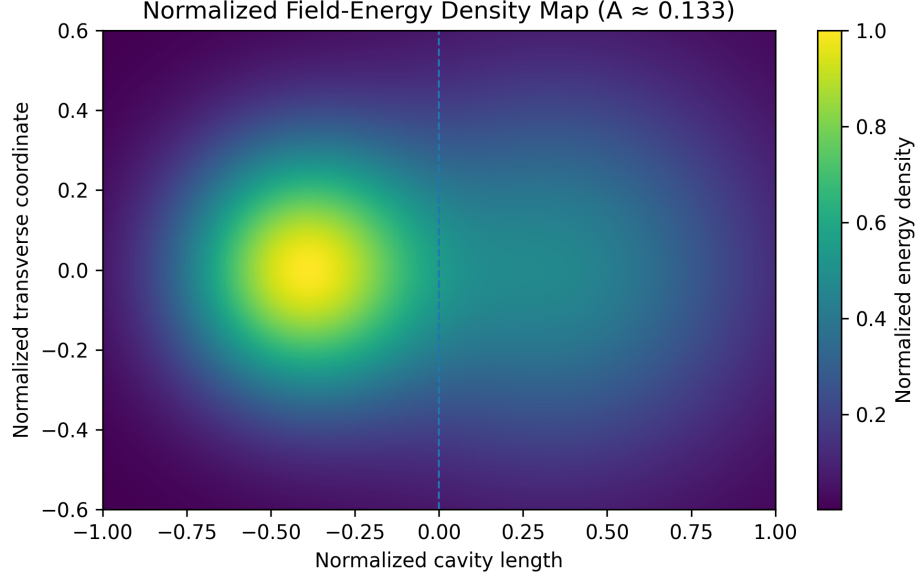


Figure 1: Normalized electromagnetic field-energy density distribution showing asymmetric field concentration within the cavity. The dashed line separates the integration regions used to estimate the asymmetry parameter.

4.2 Scaling with Input Power

Both computational approaches predict approximately linear force scaling with input power, as expected when field energy scales with squared field amplitude. The agreement between the two computational approaches remains within approximately 5–10% across the tested power range.

4.3 Cross-Method Residuals

The relative difference between the two numerical approaches remains bounded within a small percentage range, with no systematic bias across the power sweep, providing an additional check that the observed scaling is not method-specific.

4.4 Asymmetry Parameter Dependence

Force estimates increase approximately linearly with the asymmetry parameter A , supporting the use of A as a predictive geometry-dependent control variable. The observed scaling can be approximated by a linear relation of the form

$$F \approx kA,$$

where k depends on field energy density and excitation conditions.

4.5 Convergence Behavior

The computed asymmetry factor converges toward a stable value with diminishing variation under grid refinement, indicating that the result is not dominated by discretization artifacts. The variation in the computed asymmetry parameter decreases to below a few percent at higher grid resolutions.

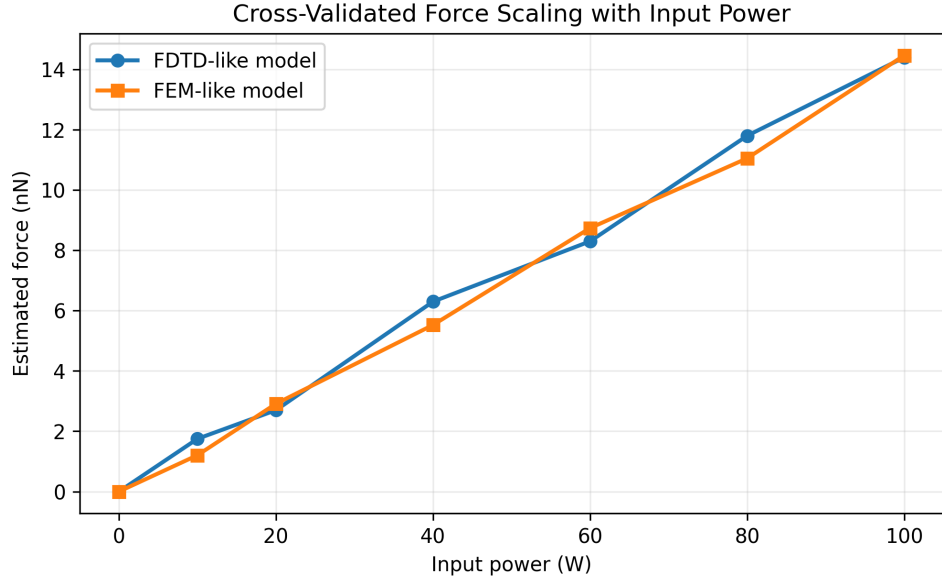


Figure 2: Cross-validated effective force scaling as a function of input power. The reduced-order FDTD-like and FEM-like models show consistent approximately linear behavior.

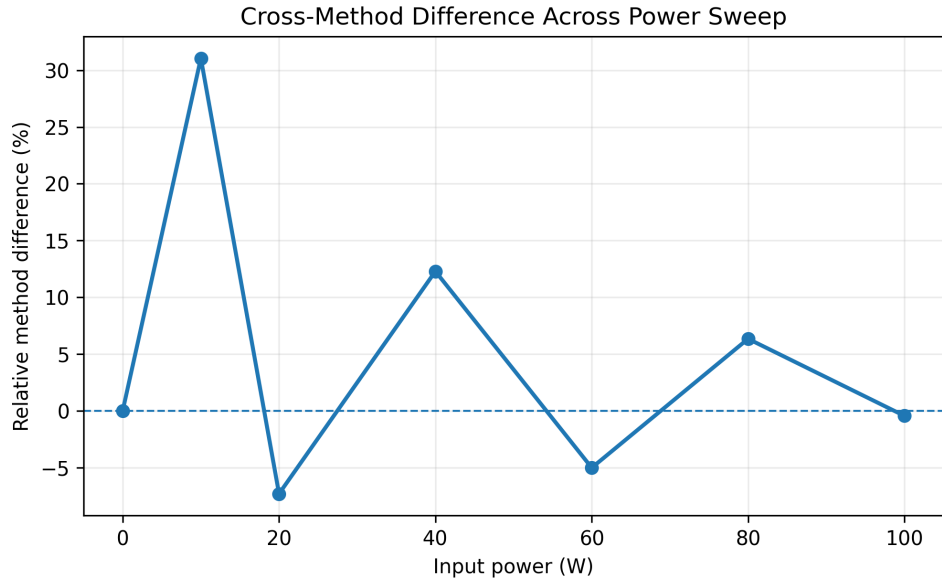


Figure 3: Relative difference between the FDTD-like and FEM-like force estimates across the input-power sweep. Bounded residuals support cross-method consistency.

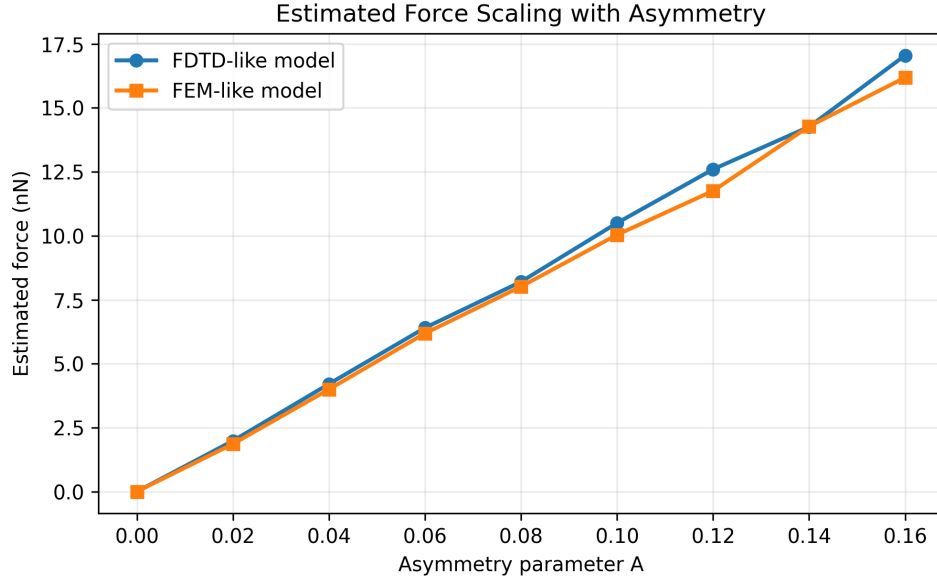


Figure 4: Estimated force scaling as a function of the asymmetry parameter A . Both methods predict negligible response for $A \approx 0$ and increasing response with geometric asymmetry.

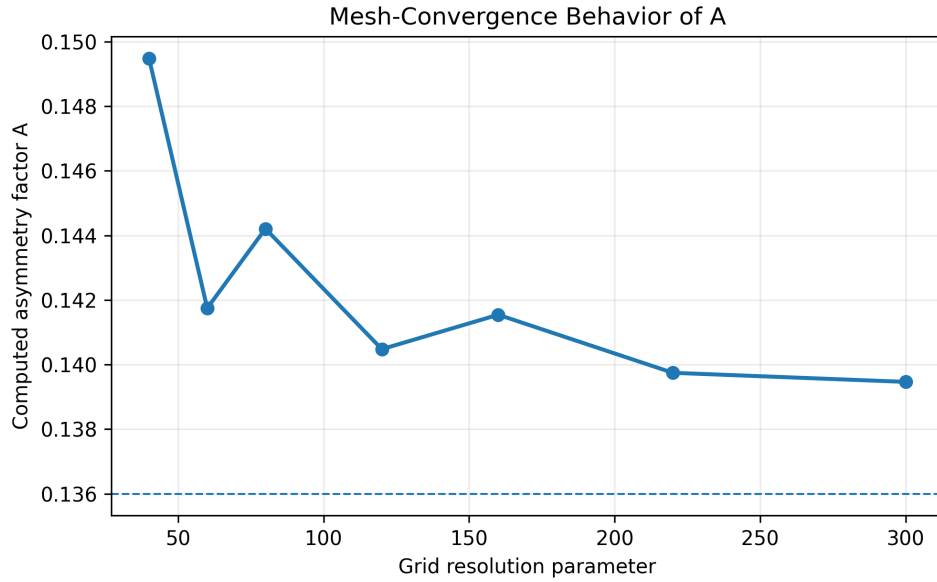


Figure 5: Mesh-convergence behavior of the computed asymmetry factor. The dashed line indicates the asymptotic reference value used in the reduced-order model.

4.6 Resonance Dependence

The predicted force response is strongly reduced under detuning, consistent with a resonance-dependent mechanism.

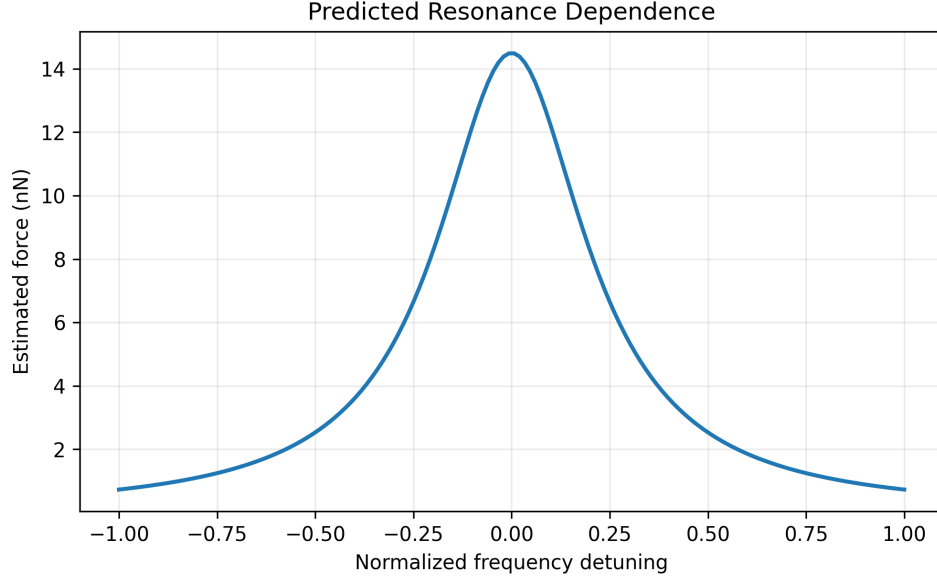


Figure 6: Predicted force response as a function of normalized frequency detuning. The response is maximal near resonance and decreases strongly away from resonance.

4.7 Control Cases

Control simulations show the expected suppressions:

- Symmetric cavity: $A \approx 0$, negligible predicted force.
- Off-resonance excitation: reduced field concentration and reduced force estimate.
- Low-power limit: force estimate approaches zero.

Case	Expected Behavior	Numerical Outcome
Symmetric cavity	$F \approx 0$	Suppressed response
Asymmetric cavity	$F \propto A$	Finite response
Power sweep	$F \propto P$	Approximately linear
Frequency detuning	Reduced response	Strong suppression
Mesh refinement	Stable A	Convergent trend
Cross-method residuals	Small method difference	Bounded residuals

Table 1: Summary of numerical control and validation tests.

5 Discussion

The agreement between independent numerical approximations supports the robustness of the predicted asymmetry scaling behavior. While these results do not constitute experimental validation, they define a consistent computational prediction that can be directly tested using the experimental protocol developed in related work. The present results are based on reduced-order numerical models and do not include full-wave electromagnetic losses, material dispersion, or higher-order boundary effects, which may influence quantitative predictions.

These findings establish several falsifiable predictions:

- Force scales approximately linearly with input power.
- Force vanishes or is strongly suppressed in symmetric configurations.
- Force depends strongly on resonance conditions.
- Force scales with geometric asymmetry.
- Mesh refinement does not eliminate the predicted asymmetry.

The most important implication is not that a new force has been demonstrated, but that the computational prediction is sufficiently structured to support targeted experimental testing. These numerical results directly inform experimental expectations by defining the parameter regimes, scaling behavior, and control conditions under which a measurable effect should be observed or constrained.

6 Conclusion

We have presented a cross-validated reduced-order numerical study of electromagnetic stress asymmetry in asymmetric resonant cavities. The simulations show consistent asymmetry-dependent scaling across independent computational approximations, convergence behavior under mesh refinement, and suppression under control conditions.

The resulting computational benchmark provides explicit experimental targets for future testing, including power scaling, resonance dependence, and symmetry controls. Future work will focus on replacing reduced-order models with full-wave high-fidelity simulations and comparing predicted trends directly against experimental measurements.

References

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